INSTITUTE OF TECHNOLOGY OF CAMBODIA

Department of Applied Mathematics and Statistics

Word Error Rate (WER)

How Wrong Is the Transcript — in One Number

Group 8

Topic : Word Error Rate (WER)

Course : Advanced Programming LLM Performance Evaluation

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1 INTRODUCTION

Why Word Error Rate Matters

One wrong word can change meaning: “turn off” vs “turn of”.

1. **Speech to text is everywhere.**
   Voice assistants, subtitles, and call transcripts.

2. **“Sounds good” is not a measurement.**
   We need one number to compare systems.

3. **WER is that number.**
   The standard score for speech recognition.

2 THEORY

What Word Error Rate Counts

Only three kinds of mistakes — and WER counts them all.

S

Substitution

Wrong word
in place of the right one

`sits → sit`

D

Deletion

A word is
missing

`the → –`

I

Insertion

An extra word
appears

`– → today`

How it works: align the two texts word by word — then count the edits. That count is the Levenshtein distance, at word level.

2 THEORY

The Formula

Three steps: align, count, divide.

$$\mathbf{WER = (S + D + I) \div N}$$

S = substitutions D = deletions I = insertions

N = words in the reference (the correct text)

Lower is better • 0% = perfect • WER can pass 100%

1 Align

Match the two texts.

2 Count

Add up S + D + I.

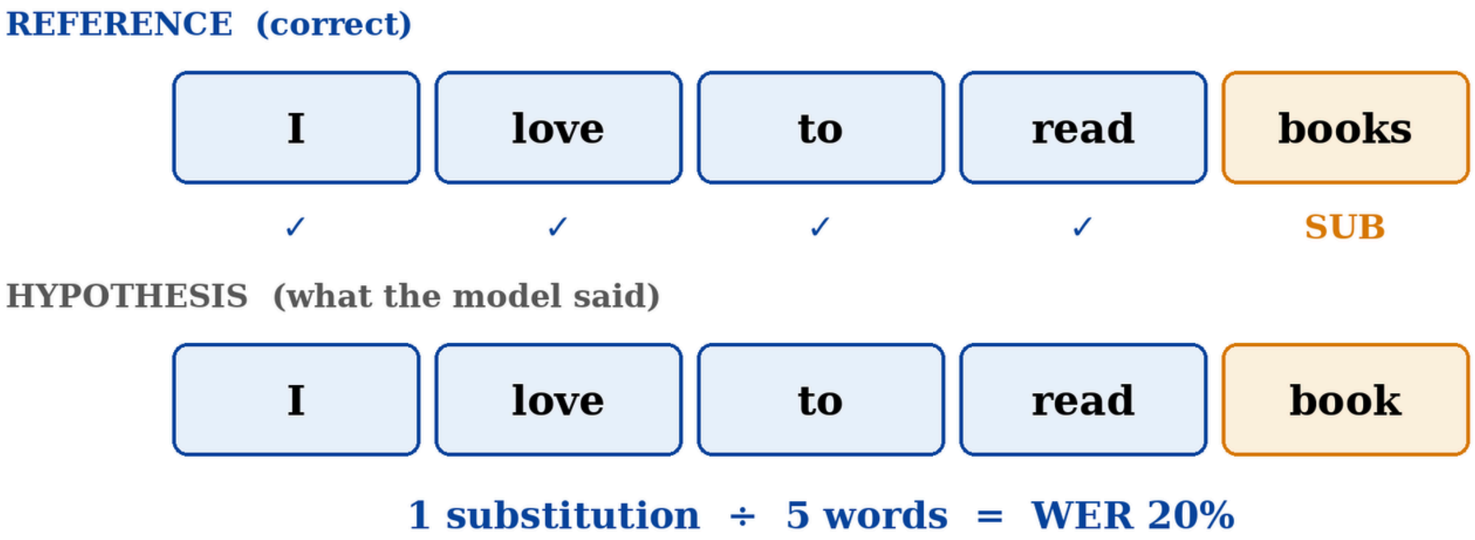
3 Divide

Divide by N.

3 EXAMPLES AND PYTHON IMPLEMENTATION

Example 1: One Substitution

Ref: "I love to read books" • Hyp: "I love to read book"

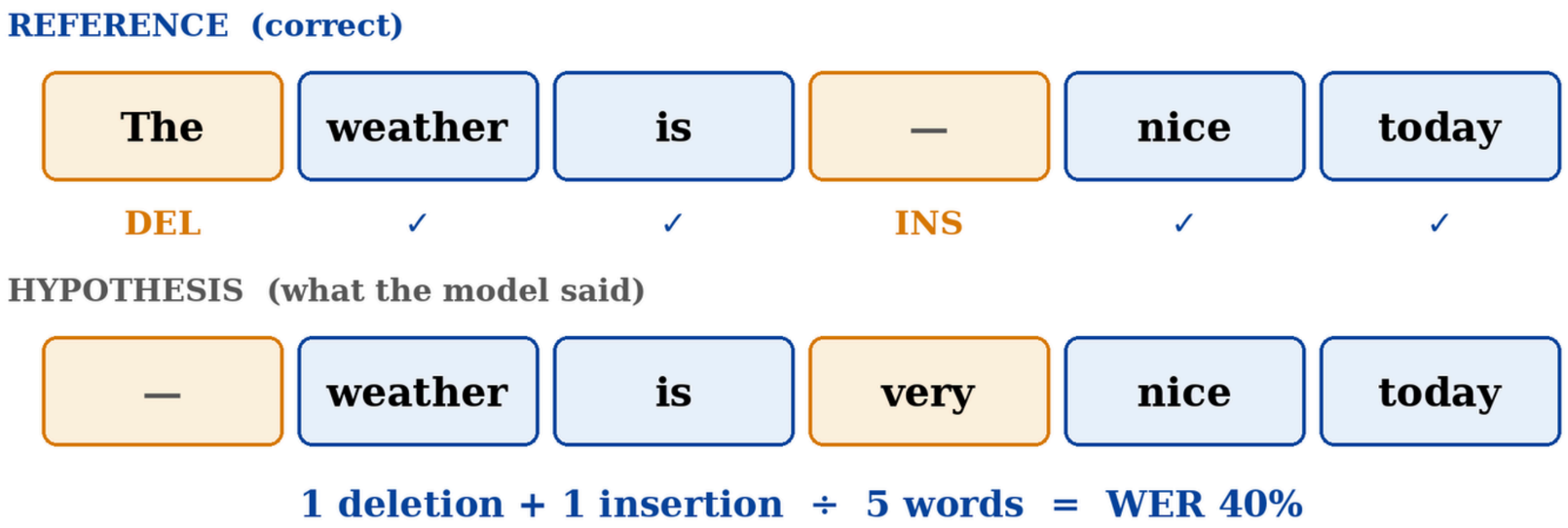


Takeaway: "books" → "book" is one word wrong — but the meaning mostly survives. WER counts words, not meaning.

3 EXAMPLES AND PYTHON IMPLEMENTATION

Example 2: Two Errors, Small Sentence

Ref: "The weather is nice today" • Hyp: "weather is very nice today"



Takeaway: Short sentences punish errors hard: with N = 5, every error is 20%.

3 EXAMPLES AND PYTHON IMPLEMENTATION

Python Implementation

Three lines with the jiwer library — the same answer as counting by hand.

```
from jiwer import wer

ref = "the cat sits on the mat"
hyp = "the cat sit on the mat"

wer_score = wer(ref, hyp)
print(round(wer_score * 100, 1), "%")
# 16.7 %
```

1. **pip install jiwer** one small library.
2. **jiwer does the alignment** word-level Levenshtein.
3. **Same answer as by hand:** 1 ÷ 6 = 16.7%.

Live demo: change one word in the hypothesis and watch WER move.

4 EXPERIMENTS AND RESULTS

Experiment: Two Models, Same Test

Fair test: only the model changes — the sentences and the scoring stay the same.

1 **Model A — Whisper-base**

Larger model. Expect fewer errors.

2 **Model B — Whisper-tiny**

Smaller and faster. Expect more errors.

3 **Same 5 sentences**

32 reference words. Same audio, same text.

4 **Same scoring**

Align, count S + D + I; WER = errors ÷ 32.

Honest note: a small class test (5 sentences). It shows the method clearly — it is not a large benchmark.

4 EXPERIMENTS AND RESULTS

Results: Whisper-Base Wins

Whisper-base made half as many errors as Whisper-tiny.

#	Reference sentence	A: Whisper-base	B: Whisper-tiny
1	the cat sits on the mat	0 errors	1 sub
2	please turn off the lights	1 del	1 sub
3	i want to book a flight to siem reap	1 ins	1 del
4	she sells seashells by the seashore	0 errors	1 sub + 1 ins
5	the meeting starts at nine o'clock	1 del	1 sub
TOTAL — 32 words		3 errors → 9.4%	6 errors → 18.8%

9.4%
Whisper-base (A) • 3 errors • accuracy 90.6%

18.8%
Whisper-tiny (B) • 6 errors • accuracy 81.2%

5 ADVANTAGES AND CONCLUSION

Advantages and Limitations

WER counts words, not understanding.

ADVANTAGES

- Simple — one number.
- Easy to compare systems.
- Shows the error type (S / D / I).
- The standard in speech recognition.

LIMITATIONS

- Ignores meaning — “book” vs “dog” count the same.
- Penalises synonyms and word order.
- Needs an exact reference text.
- Small tests are noisy.

5 ADVANTAGES AND CONCLUSION

Conclusion

Three things to remember.

1. **WER = (S + D + I) ÷ N**
   Lower is better. 0% is perfect.

2. **In our test, base wins**
   Whisper-base 9.4% vs Whisper-tiny 18.8%.

3. **Count words — then listen**
   Pair WER with human listening for meaning.

When to use it: WER for speech — translation → BLEU, summaries → ROUGE, meaning → BERTScore.

References: jiwer documentation • Radford et al. (2022), Whisper • Course assignment brief, LLM Performance Evaluation

THANK YOU FOR YOUR LISTENING

Questions and Answers